Cristina López Alcázar

Computer Engineering & Business Administration · AI for Healthcare

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RESEARCH INTERESTS

Machine learning and deep learning applied to medical imaging, clinical decision support and computational biology, with a focus on building reproducible, deployable systems that bridge research- grade models and real clinical impact. Particularly interested in explainability, uncertainty quantification and topology- aware methods for healthcare AI.

EDUCATION

Universidad Carlos III de Madrid *Sep 2021 – Jun 2026*

Double Degree in Computer Science and Engineering & Business Administration · Grade: 8.61 / 10

- **Relevant coursework - CS/AI:** Artificial Intelligence, Machine Learning, Artificial Neural Networks, Algorithm Design, Data Structures & Algorithms, Heuristics & Optimization, Language Processors, Computer Architecture, Distributed Systems, Internet of Things, Interactive & Ubiquitous Systems, User Interfaces, Software Engineering, Software Project Management, Cybersecurity & Cryptography, Databases, Operating Systems, Computer Networks.
- **Relevant coursework - Math & Stats:** Linear Algebra, Calculus, Differential Calculus, Statistics I & II, Discrete Mathematics, Logic, Econometrics.
- **High Honor Distinctions** (Matrícula de Honor) awarded in 7 subjects across both degrees.

Texas A&M University - International Academic Exchange *Aug 2024 – May 2025*

Coursework in Computer Science and Engineering · GPA: 3.88 / 4.00

- Selected for a competitive international exchange; full academic year completed entirely in English.

I.E.S. Los Rosales - Bachillerato de Excelencia (High School Honors) *Sep 2019 – Jun 2021*

Final Grade: 10 / 10 · Matrícula de Honor (Highest Distinction)

- Selected for a highly competitive academic program emphasizing research and advanced coursework.

AWARDS & HONORS

1st Prize - C4DX 2026 Challenge *Jun 2026*

Cátedra IRSST – Universidad Carlos III de Madrid

- Winning team project (**SpineUp**) in the 6th edition of the C4DX Challenge on Creativity for Digital Transformation in Occupational Health and Safety. Awarded among multiple competing university teams; project presented at the 6th IRSST- UC3M Workshop.

1st Place - JetBrains Hackathon (PyCharm Plugin Track) *Apr 2026*

- Built a context- aware PyCharm plugin integrating GitHub, Jira, Slack and email to generate dynamic code annotations. MVP delivered under time- constrained conditions.
- github.com/crislpzalc/HACKATHON-JETBRAINS

Amgen Scholars Program - Institut Pasteur, Paris *Apr 2026*

- Selected for one of the most competitive international undergraduate research programs in life sciences (acceptance rate <5%).

Excellence Scholarship of the Community of Madrid *Awarded for 2024–2025 academic year*

- Granted by the regional government to top- performing university students based on academic merit.

First- Year Tuition Waiver - Universidad Carlos III de Madrid *Sep 2021*

- Granted for outstanding academic merit at university entry.

RESEARCH EXPERIENCE

Institut Pasteur, Paris - Amgen Scholars Research Intern (Incoming) *Jul 2026 – Sep 2026*

BioImage Analysis Unit · PI: Dr. Jean- Christophe Olivo- Marin · Mentor: Dr. Julie Mabon

- Developing a deep learning pipeline for 3D vascular network extraction from confocal microscopy imaging data.
- Designing and evaluating segmentation models with a focus on preserving vascular topology and connectivity.
- Building an end- to- end pipeline (raw volume → segmentation → skeleton → graph) for quantitative biological analysis.
- Results to be presented at the **University of Cambridge** (Amgen Scholars Symposium, September 2026).

SpineUp - Smart Textile for Posture Reeducation in Occupational Settings *Feb 2026 – Jun 2026*

Final Project, Software Project Management - Universidad Carlos III de Madrid

- Co- led a 6- person interdisciplinary team (CS + Business) designing an IoT- based posture reeducation system combining capacitive textile sensors and pneumatic actuators.
- Architected the data flow between sensor layer, mobile application and aggregated analytics dashboard for occupational health teams.
- Applied formal project management methodology: Use Case Points estimation, iterative planning, transversal quality assurance and full requirements- to- implementation traceability.
- Awarded **1st Prize at the C4DX 2026 Challenge** (Cátedra IRSST- UC3M).

PROFESSIONAL EXPERIENCE

Siemens Rail Automation - Data Manager Intern *Oct 2023 – Apr 2024*

- Managed and restructured large- scale operational datasets in a safety- critical railway automation environment, improving data consistency and traceability across engineering teams.
- Cleaned, standardized and validated heterogeneous data sources, establishing a more reliable structure for downstream analysis.
- Collaborated with multidisciplinary engineering teams, gaining hands- on experience with data reliability standards in regulated industrial contexts.

Peer Mentor & Onboarding Advisor - UC3M *Sep 2022 – Jun 2024*

- Selected to mentor 6 first- year students across two academic years, providing academic guidance and support during their transition to university life.
- Developed communication and coaching skills in a structured peer- mentoring program.

SELECTED PROJECTS IN HEALTHCARE AI

EEG Seizure Detection System (Clinical Decision Support System) *Ongoing*

- End- to- end system for seizure detection from EEG signals using the CHB- MIT dataset (BIDS format).
- Modular pipeline: data ingestion, preprocessing, feature extraction, 1D CNN training, evaluation and deployment- oriented inference.
- Advanced clinical- grade components: model calibration, uncertainty estimation and explainability.
- REST API (FastAPI) for model inference, with reproducible Docker- based workflows.
- *github.com/crislpzalc/neuro- eeg- cdss*

Multi- label Chest X- ray Classification (NIH Dataset) *2025*

- End- to- end PyTorch pipeline for multi- label classification of 15 thoracic findings from 112k+ chest X- ray images.
- Multi- label stratified splitting, custom dataset loaders, transfer learning with ResNet- 18.
- Addressed strong class imbalance with weighted loss functions and per- class threshold optimization. Reached **macro AUC- ROC of 0.821.**
- kaggle.com/cristinalopezalcazar/multi-label-chest-x-ray-classification-nih-chest

Pneumonia Detection from Chest X- ray Images *2025*

- Full end- to- end pipeline on 6,298 chest X- ray images (Kaggle). Compact CNN (11.3M params) with BatchNorm, Dropout and threshold tuning.
- Achieved **96.8% accuracy, 0.97 F1- score and 0.99 AUC- ROC** on the test set.
- Implemented Grad- CAM visualisations for clinical explainability; live web demo deployed for open- access reproducibility.
- *github.com/crislpzalc/Pneumonia_detection_CNN*

TECHNICAL SKILLS

Programming Languages: Python (advanced), Java, C, C++, JavaScript, SQL, HTML/CSS, Bash

Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit- learn, computer vision, medical imaging, transfer learning, Grad- CAM, model calibration & uncertainty estimation

ML Systems & MLOps: End- to- end pipelines, data preprocessing, evaluation on imbalanced datasets, FastAPI, Docker & Docker Compose, Hugging Face Spaces, TensorBoard, Linux

Cloud & Software Engineering: Google Cloud Platform (GCP), modular design, REST APIs, system integration, testing & debugging, version control (Git)

Scientific Tools: NumPy, Pandas, Matplotlib, Jupyter

LANGUAGES

Spanish: Native

English: Full professional proficiency · TOEFL iBT 100/120